EXCEL

POWER QUERY · PIVOT

COURSE SLIDES · WSQ

# Excel Power Query and Power Pivot

WSQ Course Code: TGS-2026064177

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

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COURSE ADMIN

# Digital Attendance (Mandatory)

### 1 AM Attendance

Scan at the start of each training day.

### 2 PM Attendance

Scan after the lunch break.

### 3 Assessment Attendance

Scan before the final assessment begins.

### 4 75% Minimum

Attendance below 75% forfeits funding and certification.

The trainer will display the SSG digital attendance QR code — scan it with your phone camera and submit.


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# About the Trainer




## Your Trainer

WSQ Certified Trainer & Assessor
(ACTA / ACLP)

### Qualifications

Degree/Masters in a data or engineering discipline; ACLP-certified.

### Industry Experience

Hands-on practitioner in data analysis and business reporting.

### Teaching Focus

Practical, lab-driven delivery — you build every artefact yourself.

### Contact

enquiry@tertiaryinfotech.com · +65 6100 0613


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# About the Trainer




## Dr Alfred Ang

Principal Trainer, Tertiary Infotech Academy

### Qualifications

PhD; ACLP-certified WSQ trainer and assessor.

### Industry Experience

20+ years across data analytics, automation and software engineering.

### Specialisation

Excel Power Query / Power Pivot, Power BI, Python analytics.

### Contact

enquiry@tertiaryinfotech.com · +65 6100 0613


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# Let's Know Each Other

1. **Your name**
   and the organisation you are from

2. **Your role**
   and how much Excel you use day to day

3. **Your data**
   the reports or files you clean up most often

4. **Your goal**
   what you want to stop doing manually after this course


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# Ground Rules

1. **Phones on silent**
   Step out quietly for calls
2. **Participate actively**
   No question is a stupid question
3. **Mutual respect**
   Agree to disagree
4. **One conversation**
   One person speaks at a time
5. **Be punctual**
   Return from breaks on time
6. **75% attendance**
   Required for funding and certification


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# What You Need for This Course

1. **Microsoft Excel for Windows**
   Microsoft 365, 2021, 2019 or 2016

2. **Power Query**
   Built in under Data > Get & Transform Data

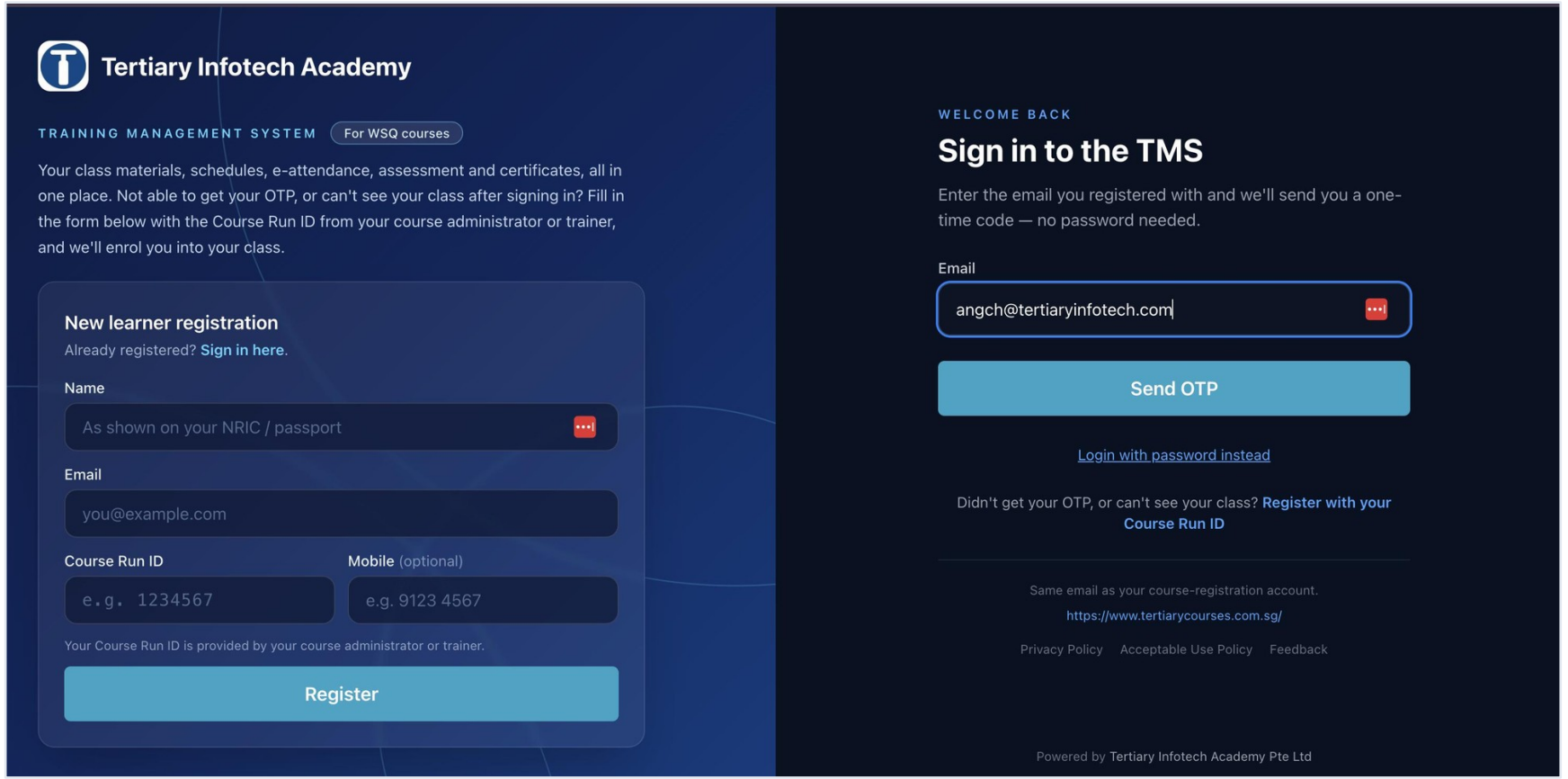
3. **Power Pivot**
   File > Options > Add-ins > COM Add-ins > enable

4. **Lab files**
   One folder per lab, downloaded from the LMS


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COURSE PORTAL

# Login to LMS and Download Course Material




Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.


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COURSE ADMIN

# Download Your Course Material

https://lms-tms.tertiaryinfotech.com

**Excel Power Query and Power Pivot**

- Trainer Slides (PPT)
- Learner Guide (PDF)
- Lesson Plan (PDF)
- Activities / Lab Files

**1 Log in**

Use the account e-mailed to you before the course.

**2 Open this course**

Excel Power Query and Power Pivot.

**3 Download the files**

Slides, Learner Guide, Lesson Plan and lab workbooks.

**4 Save locally**

Unzip the labs folder to your Desktop before Lab 01.

All courseware is on the LMS/TMS portal at lms-tms.tertiaryinfotech.com — download it before the first lab.


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# Skills Framework Alignment

1

**TSC Title**

Data and Statistical Analytics

2

**TSC Code**

EGS-BIN-4055-1.1

3

**Proficiency Level**

Level 4 — apply and adapt analysis methods

4

**Assessment**

Written Assessment + Practical Performance


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# TSC Knowledge and Abilities Assessed

| Code | What it covers | Where it is taught |
|---|---|---|
| K1 | Operations of data transformation techniques | Topic 1, 2 · Labs 1–13 |
| K2 | Conditions required for each transformation and join | Topic 1–3 · Labs 3–19 |
| K3 | Interpretation of query and data-model results | Topic 2, 3 · Labs 12–19 |
| K4 | Types of data analysis software and capabilities | Topic 1, 2 · Labs 1, 2, 13 |
| A1 | Review data sets to uncover trends or patterns | Topic 1 · Labs 1–6 |
| A2 | Develop methods for complex data sets | Topic 2 · Labs 7–13 |
| A3 | Apply analytics to examine business issues | Topic 3 · Labs 14–19 |

**WHEN IT MATTERS** Every K and A code is taught in a topic AND practised in a lab — nothing is assessed that you have not built.


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# Learning Outcomes

## 1 LO1 · Review

Use Power Query to load, clean and reshape data sets so trends and patterns become visible.

## 2 LO2 · Develop

Build methods in Power Query and Power Pivot — logic, dates, grouping and DAX measures — for complex data.

## 3 LO3 · Apply

Apply advanced joins, approximate and fuzzy matching to examine real business issues.

### HOW YOU WILL GET THERE

Each outcome is taught as a concept, demonstrated by the trainer, then built by you in the labs — 19 hands-on labs across the two days.


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TOPIC 1 · 35% OF THE COURSE

# Course Outline — Topic 1

### 1 Lab 01

Load Sales Data and Fix Data Types

### 2 Lab 02

Combine an External Workbook as a Connection Only

### 3 Lab 03

Fill Down and Fill Up a Ragged Report

### 4 Lab 04

Split Columns by Delimiter and into Rows

### 5 Lab 05

Add Conditional, Custom and Example-Driven Columns

### 6 Lab 06

Merge Columns, Sort and Filter

**REVIEW DATA WITH POWER QUERY**

Data types · Getting data · Shaping columns · Sorting and filtering


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TOPIC 2 · 35% OF THE COURSE

# Course Outline — Topic 2

**1 Lab 07**

Conditional Logic with IF in M

**2 Lab 08**

Nest IF with AND for a Priority Flag

**3 Lab 09**

Calculate Deadlines with Date.AddDays

**4 Lab 10**

Unpivot a Wide Report into Tidy Rows

**5 Lab 11**

Pivot, Pivot Without Aggregation, and Append

**6 Lab 12**

Summarise with Group By

**7 Lab 13**

Build a Power Pivot Data Model with DAX Measures

**DATA ANALYSIS WITH POWER QUERY AND POWER PIVOT**

Logic · Dates · Pivot and unpivot · Group By · Append · The Data Model


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TOPIC 3 · 30% OF THE COURSE

# Course Outline — Topic 3

1 **Lab 14**

Walk Through All Six Join Kinds

2 **Lab 15**

Left Outer Join versus XLOOKUP

3 **Lab 16**

Merge on Multiple Fields

4 **Lab 17**

Approximate Match: Binning and Conditional Columns

5 **Lab 18**

Cross Join and Drill Down to a Variable

6 **Lab 19**

Fuzzy Matching and Transformation Tables

**ADVANCED DATA ANALYSIS USING POWER QUERY**

The six joins · Merge · Approximate match · Fuzzy matching · Transformation tables


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COURSE ADMIN

# Lesson Plan — Day 1

| Time | Session | Detail |
|---|---|---|
| **9:30 – 9:45** | Digital Attendance (AM) | Scan the SSG QR code |
| **9:45 – 10:15** | Introductions & Course Overview | Outcomes, outline, assessment |
| **10:15 – 12:30** | Topic 1 (Part 1) | Power Query concepts · Labs 01–03 |
| **12:30 – 13:30** | Lunch Break | — |
| **13:30 – 13:40** | Digital Attendance (PM) | Scan the SSG QR code |
| **13:40 – 15:45** | Topic 1 (Part 2) | Column shaping · Labs 04–06 |
| **15:45 – 18:00** | Topic 2 (Part 1) | Logic and dates · Labs 07–09 |
| **18:00 – 18:30** | Day 1 Recap | Q&A and consolidation |


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# Lesson Plan — Day 2

| Time | Session | Detail |
|---|---|---|
| **9:30 – 9:40** | Digital Attendance (AM) | Scan the SSG QR code |
| **9:40 – 12:30** | Topic 2 (Part 2) | Pivot, Group By, Power Pivot · Labs 10–13 |
| **12:30 – 13:30** | Lunch Break | — |
| **13:30 – 13:40** | Digital Attendance (PM) | Scan the SSG QR code |
| **13:40 – 16:00** | Topic 3 | Joins and matching · Labs 14–19 |
| **16:00 – 16:30** | Revision | Consolidation and Q&A |
| **16:30 – 16:45** | Course Feedback & TRAQOM | Mandatory survey |
| **16:45 – 18:30** | Digital Attendance + Final Assessment | WA 1 hr + PP 1 hr |


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# Briefing for Assessment

1. **Clear your desk**
   Phones and materials under the table or on the floor

2. **No photos or recording**
   Assessment scripts must not be captured

3. **No discussion**
   Work individually throughout

4. **Black or blue pen**
   For hard-copy scripts; no correction tape or liquid paper

5. **Scripts collected**
   All scripts are collected when time is up

6. **Open book**
   Slides, Learner Guide and approved materials only


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# Final Assessment

### 1 Written Assessment (WA)

Short-Answer Questions · 1 hour · open book. Tests your knowledge of Power Query and Power Pivot concepts.

### 2 Practical Performance (PP)

Hands-on tasks in Excel · 1 hour · open book. You perform the transformations you practised in the labs.

### 3 Result

Competent / Not Yet Competent. Both instruments must be passed.

### 4 Re-assessment

One re-assessment opportunity is offered if assessed Not Yet Competent.

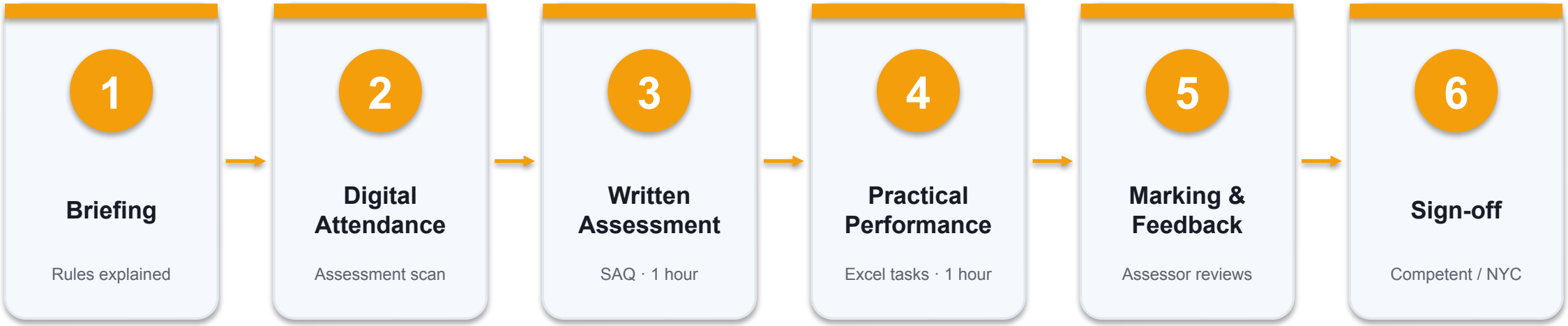
**OPEN BOOK MEANS**

You may refer to these slides, the Learner Guide and any approved course material. You may not use the internet, AI assistants or discuss with others.


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# Assessment Flow




**SIGN-OFF**

The assessor records the outcome, you sign the assessment record, and results are released through the LMS/TMS portal.


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# Criteria for Funding

1

## 75% attendance

Based on the SSG digital attendance record

2

## Assessed Competent

Both the WA and the PP must be passed

3

## TRAQOM survey

Mandatory course feedback survey

4

## Certificate

Certificate of Achievement issued on completion


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01

TOPIC 1

# Review Data with Power Query

Data types · Getting data · Shaping columns · Sorting and filtering


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TOPIC 1 · WHY POWER QUERY

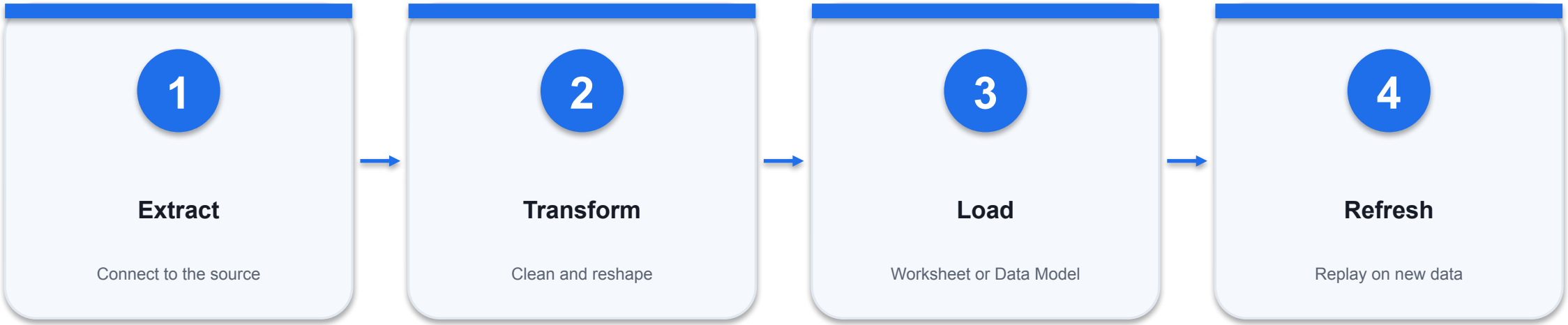
# Stop repeating the same clean-up every month.

Power Query records your transformation once and replays it on every refresh.


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TOPIC 1 · THE ETL LOOP

# What Power Query Actually Does




**THE POINT**

You build the recipe ONCE. Next month you replace the source file, click Refresh, and every step runs again in the same order.


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# Worksheet Formulas vs Power Query

## ✓ Power Query

- Steps are recorded and re-run on Refresh
- Handles millions of rows outside the grid
- The whole clean-up is auditable in Applied Steps
- New rows are picked up automatically
- One query feeds many reports

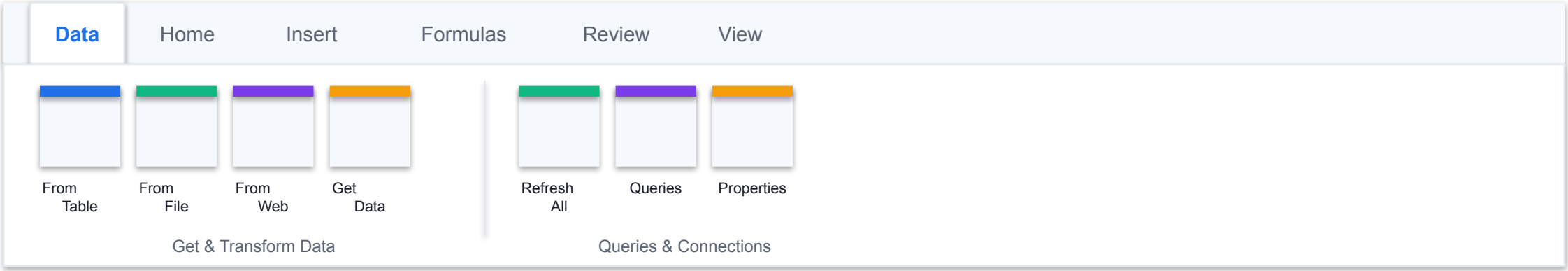
## ✗ Manual formulas / copy-paste

- Must be re-done or dragged down every month
- Slows the workbook down as rows grow
- No record of what was changed, or why
- Breaks silently when the layout shifts
- Each analyst rebuilds their own version

Rule of thumb: if you have done it twice by hand, it belongs in Power Query.


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# Where Power Query Lives in Excel




### Data > From Table/Range

Loads the selected worksheet table into the Power Query Editor.

### Data > Get Data

Connects to an external workbook, CSV, database, folder or web source.

### Queries & Connections

The side pane listing every query, its load state and its row count.


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# The Power Query Editor — Five Areas You Must Know

1

### 1. The Ribbon

Home, Transform, Add Column and View tabs

2

### 2. Queries Pane

Every query in the workbook, on the left

3

### 3. Data Preview

The first 1000 rows — a preview, not the full data

4

### 4. Query Settings

Properties and the Applied Steps list

5

### 5. Status Bar

Column count, row count and profiling info

6

### Applied Steps

The heart of it — your recorded, editable recipe


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# Applied Steps — Your Audit Trail

**Every click you make becomes a step you can rename, reorder, edit or delete.**

### Order matters

Steps run top to bottom. Filtering before a merge is fast; after it is slow.

### Everything is editable

Click the gear icon on a step to reopen the dialog you used.

### Delete with care

Removing an early step can break the ones that follow it.


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# Data Types — Set Them in the Query, Not the Grid

| Type | What it holds | Common failure |
|---|---|---|
| **Text** | Names, codes, IDs, postal codes | Leading zeros lost if typed as number |
| **Whole Number** | Counts and quantities | Will not sum while stored as text |
| **Decimal Number** | Prices, amounts, rates | Currency prefix blocks the type change |
| **Date** | Calendar dates | dd/mm/yyyy read as US format — use Using Locale |
| **Date/Time** | Timestamps | Time zone shifts the date by a day |
| **True/False** | Logical flags | The words TRUE/FALSE stored as text |
| **Any** | Type not yet set | Excel guesses — and eventually guesses wrong |

**WHEN IT MATTERS** A wrong data type is the single most common cause of a broken refresh. Set every column's type explicitly as the FIRST thing you do.


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# Which Load Option Should I Choose?




### Load To > Table

Loads to a worksheet. Use for the final, report-ready output only.

### Only Create Connection

Keeps it out of the grid. Use for staging queries that feed a merge or append.

Tick 'Add this data to the Data Model' as well when the table will be used in Power Pivot. Loading everything to worksheets is the fastest way to bloat a workbook.


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# Getting Data — The Sources You Will Use

### 1 From Table/Range

The data already on a worksheet. The starting point for Lab 01.

### 2 From Workbook

Another Excel file — the pattern in Lab 02 for the monthly price list.

### 3 From Text/CSV

Exports from any system that can write a delimited file.

### 4 From Folder

Every file in a folder combined into one table — the monthly-file pattern.

**THE CONNECTION IS LIVE**

Power Query stores the path, not a copy. Replace the source file with a newer one of the same shape and Refresh brings the new data straight through.


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TOPIC 1 · HANDS-ON LAB

LAB 01 · 30 MIN

# Lab 01: Load Sales Data and Fix Data Types

The Southbridge Trading sales extract arrives with everything stored as text: order dates read as words, amounts carry a stray currency prefix, and quantities will not sum.

**YOU'LL PRODUCE**

A clean Sales query loaded to a worksheet with every column correctly typed.

**HOW YOU'LL KNOW IT WORKED**

Quantity and Amount sum in the status bar, Order Date filters offer date pickers, and Column quality reports 0% error on every column.

**MAPS TO** LO1 · A1 · K1, K4 — Query data from a table or range and set correct data types.

LAB 01

**lab-01-load-and-data-types.xlsx**

Excel, Power Query Editor, Data Profiling · Full step-by-step: Learner Guide, Lab 01


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LAB 02 · 30 MIN

# Lab 02: Combine an External Workbook as a Connection Only

The regional price list lives in a separate workbook that the finance team refreshes monthly. The learner connects to that external file, loads it as Connection Only so it does not clutter the workbook grid, and confirms the connection refreshes independently.

**YOU'LL PRODUCE**

A PriceList staging query held as Connection Only, ready to be merged later.

**HOW YOU'LL KNOW IT WORKED**

The Queries & Connections pane lists PriceList as 'Connection only', and no new worksheet was created.

**MAPS TO** LO1 · A1 · K4 — Query data from another Excel file and load as connection only.

LAB 02

**lab-02-external-connection.xlsx**

Excel, Power Query Editor, External workbook source · Full step-by-step: Learner Guide, Lab 02


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TOPIC 1 · FILL

# Fill Down and Fill Up

**Reports written for humans leave gaps. Fill turns them back into real data.**

### Fill Down

Carries a value INTO the blank rows beneath it — the classic merged-cell repair.

### Fill Up

Carries a value into the blanks ABOVE it, for labels that sit under their group.

### Watch for null vs blank

Fill only replaces null. Cells holding an empty string need Replace Values first.


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LAB 03 · 30 MIN

# Lab 03: Fill Down and Fill Up a Ragged Report

A departmental headcount report was typed for human eyes: the region name appears only on the first row of each block, and the reporting quarter sits below its group. The learner uses Fill Down and Fill Up to push those values across every row so the table becomes machine-readable.

**YOU'LL PRODUCE**

A rectangular headcount table with no blank grouping cells.

**HOW YOU'LL KNOW IT WORKED**

Every row carries a Region and a Quarter value, and the row count matches the number of real headcount records.

**MAPS TO** LO1 · A1 · K2 — Repair merged-cell style gaps with Fill Down and Fill Up.

LAB 03

**lab-03-fill-down-up.xlsx**

Excel, Power Query Editor, Fill Down / Fill Up · Full step-by-step: Learner Guide, Lab 03


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TOPIC 1 · SPLIT COLUMN

# Splitting a Column — Four Ways

| Method | What it does | Use it when |
|---|---|---|
| **By Delimiter** | Cuts at a chosen character | Names, codes, comma-separated lists |
| **By Number of Characters** | Cuts at a fixed width | Fixed-width legacy exports |
| **By Positions** | Cuts at named positions | Mainframe or COBOL-style records |
| **By Lowercase to Uppercase** | Cuts at a case change | camelCase or PascalCase fields |

**WHEN IT MATTERS** Split into COLUMNS makes the row wider. Split into ROWS makes the table taller — that is how you normalise a comma-separated list into one record per value.


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# Split Into Columns vs Split Into Rows

## ✓ Split into Rows

- One product per row — a tidy, analysable table
- Row count grows; column count stays fixed
- PivotTables and merges work directly on it
- The right choice for a repeating list

## ✗ Split into Columns

- Product 1, Product 2, Product 3 … columns
- Breaks the moment someone buys a fourth item
- Every downstream formula must be rewritten
- Only correct for a FIXED number of parts (e.g. first/last name)

Ask: can the number of parts change? If yes, split into rows.


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TOPIC 1 · HANDS-ON LAB

LAB 04 · 30 MIN

# Lab 04: Split Columns by Delimiter and into Rows

The CRM export packs the customer's full name into one field and lists every product a customer bought as a comma-separated string. The learner splits the name into First and Last Name columns, then splits the product list into rows so each purchase becomes its own record.

**YOU'LL PRODUCE**

A normalised customer table with one product per row.

**HOW YOU'LL KNOW IT WORKED**

A customer who bought three products now occupies three rows, and no product value has a leading space.

**MAPS TO** LO1 · A1 · K2 — Split a column by delimiter, into columns and into rows.

LAB 04

**lab-04-split-columns.xlsx**

Excel, Power Query Editor, Split Column · Full step-by-step: Learner Guide, Lab 04


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# Three Ways to Add a Column

### 1 Conditional Column

A guided dialog that writes if/then/else for you. Best for banding and classification rules.

### 2 Custom Column

You write the M expression yourself. Needed for anything the dialog cannot express.

### 3 Column From Examples

You type the answer you want on the first row and Power Query infers the pattern.

### 4 Index Column

A sequential row number — useful for ranking and for restoring an original order after sorting.

**WHICH ONE FIRST?**

Try the Conditional Column dialog first, then Column From Examples, and drop to a Custom Column only when neither can express the rule. All three produce the same kind of Applied Step.


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# Conditional Column — What the Dialog Writes

The Value Band rule from Lab 05, as the dialog builds it in M.

```
M

if [Amount] >= 5000 then "High"
else if [Amount] >= 1000 then "Medium"
else "Low"
```

**Order is everything**

The FIRST matching rule wins. Put the highest threshold at the top.

**Test the boundary**

Amount exactly 5000 is High because the test is >=, not >.

**else is compulsory**

Every if needs a final else, or the query returns an error.

**Set the type after**

A new column arrives as Any — set it to Text or Decimal explicitly.


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LAB 05 · 30 MIN

# Lab 05: Add Conditional, Custom and Example-Driven Columns

Management wants each order classified into a value band, a line total calculated, and an order reference built from existing fields.

**YOU'LL PRODUCE**

A Sales query carrying Value Band, Line Total and Order Ref columns.

**HOW YOU'LL KNOW IT WORKED**

Value Band shows High/Medium/Low correctly at the 1000 and 5000 boundaries, and Line Total equals Quantity x Unit Price on every row.

**MAPS TO** LO1 · A1 · K1, K2 — Add conditional columns, custom columns and Column From Examples.

LAB 05

**lab-05-add-columns.xlsx**

Excel, Power Query Editor, Conditional Column, Custom Column, Column From Examples · Full step-by-step: Learner Guide, Lab 05


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# Sort and Filter INSIDE the Query

**Filtering at source means Excel never loads the rows you did not want.**

### Filter early

A filter placed before a merge or group is dramatically faster than one after.

### Filters are permanent

Unlike a worksheet AutoFilter, a query filter removes rows from the result.

### Sort is a step too

Sorting by two columns records two steps — check the order in Applied Steps.


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LAB 06 · 30 MIN

# Lab 06: Merge Columns, Sort and Filter

The despatch team needs a single address label field and a shortlist of the orders that actually need attention. The learner merges the address parts into one column with a chosen separator, then applies sorting and filtering inside the query so the refresh always returns the shortlist.

**YOU'LL PRODUCE**

A despatch shortlist query, sorted and filtered at source.

**HOW YOU'LL KNOW IT WORKED**

The query returns only non-cancelled orders above 1000, newest first, and Address Label reads as one comma-separated string.

**MAPS TO** LO1 · A1 · K2, K3 — Merge columns, then sort and filter data in Power Query.

LAB 06

**lab-06-merge-sort-filter.xlsx**

Excel, Power Query Editor, Merge Columns, Sort, Filter · Full step-by-step: Learner Guide, Lab 06


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# Topic 1 Recap

1. **ETL, recorded**
   Extract, Transform, Load — replayed on every Refresh

2. **Types first**
   Set every column's data type before anything else

3. **Connection Only**
   Keep staging queries out of the worksheet grid

4. **Fill repairs gaps**
   Fill Down and Fill Up rebuild machine-readable tables

5. **Split by rows**
   When the number of parts can change

6. **Three add-column tools**
   Conditional, Custom, and From Examples


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# Lunch Break

1 hour · Digital Attendance (PM) on return


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02

TOPIC 2

# Data Analysis with Power Query and Power Pivot

Logic · Dates · Pivot and unpivot · Group By · Append · The Data Model


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TOPIC 2 · FROM CLEAN TO USEFUL

# Cleaning data is only half the job.

Now you add the logic, the dates and the aggregation that answer the question.


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# Excel Formulas vs Power Query M

| Aspect | Excel worksheet | Power Query M |
|---|---|---|
| **Syntax** | IF(A2>=500, 0, 15) | if [Amount] >= 500 then 0 else 15 |
| **Case** | IF, AND, OR — not case sensitive | if, and, or — MUST be lowercase |
| **Refers to** | Cell addresses (A2) | Column names in [square brackets] |
| **else** | Optional third argument | Compulsory — every if needs an else |
| **Runs** | Once per cell, on every calculation | Once per row, on refresh only |
| **Text join** | & or CONCAT() | & (text only — numbers need Text.From) |

**WHEN IT MATTERS** M is case sensitive throughout. 'If' or 'IF' will not work — only 'if'. This is the single most common error when writing your first custom column.


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TOPIC 2 · IF IN M

# Writing IF Logic in M

The Shipping Cost rule from Lab 07 — written by hand in a Custom Column.

```
M

if [Amount] >= 500 then 0
else 15

// a comparison alone returns true/false
[Amount] >= 500
```

**Lowercase keywords**

if, then, else — capitalising any of them is an immediate error.

**Columns in brackets**

[Amount] refers to the column. Spelling and case must match exactly.

**No else = error**

Delete the else and Power Query refuses to build the column.

**Comparisons are values**

[Amount] >= 500 on its own returns TRUE/FALSE — set the type to True/False.


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# Nesting Conditions — the AND / OR Truth Table

| Amount >= 5000 | Days Outstanding > 30 | and → | or → |
|---|---|---|---|
| **TRUE** | TRUE | TRUE (Escalate) | TRUE |
| **TRUE** | FALSE | FALSE | TRUE (Watch) |
| **FALSE** | TRUE | FALSE | TRUE (Watch) |
| **FALSE** | FALSE | FALSE | FALSE (Normal) |

**WHEN IT MATTERS** 'and' needs BOTH sides true; 'or' needs EITHER. Lab 08 builds exactly this rule — rows SO-9201 to SO-9204 cover all four lines of this table.


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LAB 07 · 30 MIN

# Lab 07: Conditional Logic with IF in M

The Conditional Column dialog cannot express every rule, so the learner moves to the Custom Column window and writes M directly. The lab contrasts Excel's IF() with M's lowercase, case-sensitive if / then / else and shows why a missing else is an error.

**YOU'LL PRODUCE**

A Shipping Cost column driven by a hand-written M if expression.

**HOW YOU'LL KNOW IT WORKED**

Orders of 500 or more show Shipping Cost 0 and Free Shipping TRUE; the boundary row at exactly 500 qualifies.

**MAPS TO** LO2 · A2 · K1, K2 — Write IF logic in the Power Query M language.

LAB 07

**lab-07-if-formulas.xlsx**

Excel, Power Query Editor, Custom Column, M language · Full step-by-step: Learner Guide, Lab 07


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TOPIC 2 · HANDS-ON LAB

LAB 08 · 30 MIN

# Lab 08: Nest IF with AND for a Priority Flag

An order is escalated only when it is both high value and already overdue. The learner nests conditions using M's and / or operators to produce a single Priority column, then verifies each branch against a worked truth table.

**YOU'LL PRODUCE**

A Priority column classifying every order as Escalate, Watch or Normal.

**HOW YOU'LL KNOW IT WORKED**

Only orders meeting BOTH the value and ageing tests read Escalate; orders meeting one read Watch.

**MAPS TO** LO2 · A2 · K2 — Nest conditions with and / or to resolve a multi-factor rule.

LAB 08

**lab-08-nested-if-and.xlsx**

Excel, Power Query Editor, Custom Column, M logical operators · Full step-by-step: Learner Guide, Lab 08


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# Date Functions You Will Actually Use

### 1 Date.AddDays

Date.AddDays([Invoice Date], 30) — the due date from payment terms.

### 2 Duration.Days

Turns the gap between two dates into a whole number of days.

### 3 Date.Year / Month

Pulls the year or month out for grouping and reporting.

### 4 Date.From

Converts text or a date/time into a clean Date value.

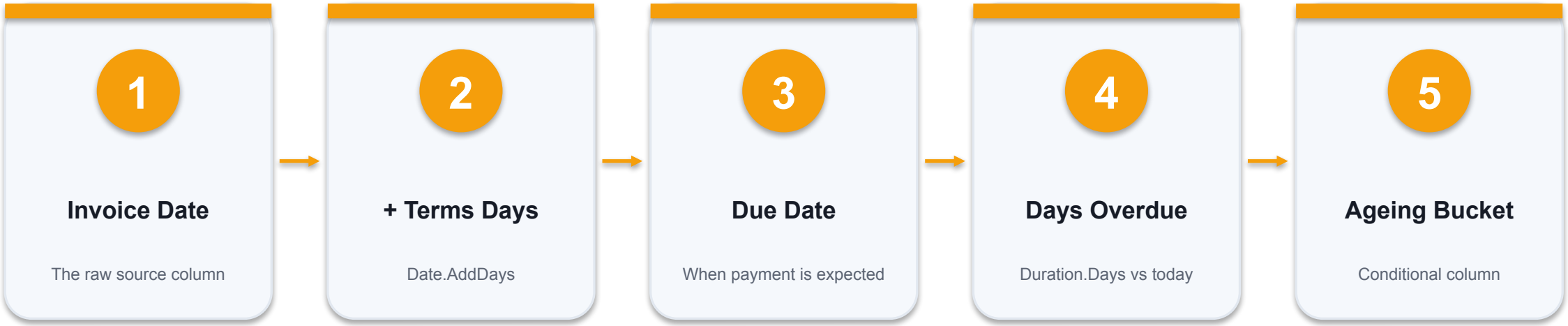
**THE WORKED EXAMPLE IN LAB 09**

Invoice INV-9001 is dated 1 May 2026 with 30-day terms. Date.AddDays gives a Due Date of 31 May 2026; against a 30 June 2026 reporting date that is exactly 30 days overdue.


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TOPIC 2 · THE DEADLINE PIPELINE

# From Invoice Date to Ageing Bucket




**WHY BUILD IT IN THE QUERY**

Every one of these columns recalculates on Refresh. Next month's invoice file drops in and the whole ageing report rebuilds itself.


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TOPIC 2 · HANDS-ON LAB

LAB 09 · 30 MIN

# Lab 09: Calculate Deadlines with Date.AddDays

Each invoice carries payment terms in days. The learner derives the due date with Date.AddDays, compares it against a fixed reporting date to produce days overdue, and bins the result into ageing buckets the finance team recognises.

**YOU'LL PRODUCE**

An invoice ageing query with Due Date, Days Overdue and Ageing Bucket.

**HOW YOU'LL KNOW IT WORKED**

An invoice dated 1 May 2026 with 30-day terms shows Due Date 31 May 2026 and Days Overdue 30.

**MAPS TO** LO2 · A2 · K1, K3 — Use Date functions to derive deadlines and ageing buckets.

LAB 09

**lab-09-adddays-deadline.xlsx**

Excel, Power Query Editor, Date functions, Custom Column · Full step-by-step: Learner Guide, Lab 09


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TOPIC 2 · UNPIVOT

# Wide is for humans. Tall is for machines.

Unpivot is the single most valuable transformation in Power Query.


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TOPIC 2 · SHAPE

# Why Unpivot Matters

## ✓ Tall / tidy — 3 columns

- Region · Month · Sales
- One row per observation
- PivotTables slice it any way you like
- A 13th month adds ROWS, not columns
- Merges and the Data Model work directly

## ✗ Wide / cross-tab — 13 columns

- Region · Jan · Feb · Mar … Dec
- One row per region, values spread across
- A PivotTable cannot group by month
- A 13th month breaks every formula
- Readable for people, useless for analysis

Lab 10 turns 5 regions x 12 month columns into 60 tidy rows.


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TOPIC 2 · UNPIVOT WARNINGS

# Three Unpivot Commands — Choose Carefully

| Command | What it unpivots | Survives a new column? |
|---|---|---|
| **Unpivot Columns** | Only the columns you selected | No — the new month is ignored |
| **Unpivot Other Columns** | Everything EXCEPT your selection | Yes — this is the safe one |
| **Unpivot Only Selected** | Only that selection, fixed forever | No — silently misses it |

**WHEN IT MATTERS** Select the columns you want to KEEP, then Unpivot Other Columns. Your query then survives next year's extra month without any edit.


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TOPIC 2 · HANDS-ON LAB

LAB 10 · 30 MIN

# Lab 10: Unpivot a Wide Report into Tidy Rows

The monthly sales report is laid out with one column per month — readable for a human, useless for a PivotTable.

**YOU'LL PRODUCE**

A tidy long-format sales table with Month and Sales columns.

**HOW YOU'LL KNOW IT WORKED**

The result has exactly three columns (Region, Month, Sales) and a row count of regions x months.

**MAPS TO** LO2 · A2 · K2, K3 — Unpivot columns, and choose between Unpivot Other Columns and Unpivot Only Selected.

LAB 10

**lab-10-unpivot.xlsx**

Excel, Power Query Editor, Unpivot · Full step-by-step: Learner Guide, Lab 10


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# Pivot — the Reverse Trip

**Pivot Column turns rows back into columns. The Don't Aggregate option is the one that matters.**

### Default: aggregate

Pivot sums or counts the values — right for numbers, wrong for text.

### Don't Aggregate

Keeps each individual value. Use it for attributes like Warranty or Supplier.

### Error if duplicated

Don't Aggregate fails when a key repeats — there would be two values for one cell.


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TOPIC 2 · APPEND

# Append — Stacking Tables

**Append adds ROWS. Merge adds COLUMNS. Confusing the two is the classic beginner error.**

### Column names must match

Exactly — including case and spaces. A mismatch creates a new column of nulls.

### Two or more tables

'Append Queries as New' keeps the sources intact and creates a third query.

### The monthly-file pattern

Append Q1 and Q2, or point From Folder at a folder of identical files.


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TOPIC 2 · HANDS-ON LAB

LAB 11 · 30 MIN

# Lab 11: Pivot, Pivot Without Aggregation, and Append

The learner reverses an unpivot to rebuild a cross-tab, meets the Don't Aggregate option that preserves individual values instead of summing them, and then appends the Q1 and Q2 extracts into a single continuous table.

**YOU'LL PRODUCE**

A pivoted cross-tab plus an appended half-year table.

**HOW YOU'LL KNOW IT WORKED**

HalfYear's row count equals Q1 rows plus Q2 rows, with no column of all nulls.

**MAPS TO** LO2 · A2 · K2, K3 — Pivot a column, use Don't Aggregate, and append queries.

LAB 11

**lab-11-pivot-append.xlsx**

Excel, Power Query Editor, Pivot Column, Append Queries · Full step-by-step: Learner Guide, Lab 11


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# Group By — Aggregating Inside the Query

### 1 Count Rows

How many orders per region — the most common aggregation.

### 2 Sum

Total sales value per group.

### 3 Average / Min / Max

Typical, smallest and largest order in each group.

### 4 All Rows

Keeps the underlying detail rows nested for later drill-through.

**BASIC VS ADVANCED**

Basic groups by one column with one aggregation. Switch to ADVANCED to group by several columns and add several aggregations at once — that is what Lab 12 does with 400 transactions.


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TOPIC 2 · HANDS-ON LAB

LAB 12 · 30 MIN

# Lab 12: Summarise with Group By

Rather than loading 20,000 transaction rows into the grid, the learner aggregates them inside the query. The lab covers basic and advanced Group By, multiple aggregations, grouping on more than one column, and keeping the underlying rows with All Rows.

**YOU'LL PRODUCE**

A regional sales summary with count, sum and average in one query.

**HOW YOU'LL KNOW IT WORKED**

Total Sales across all groups equals the grand total of the source Amount column.

**MAPS TO** LO2 · A2 · K1, K3 — Aggregate rows to a summary grain with Group By.

LAB 12

**lab-12-group-by.xlsx**

Excel, Power Query Editor, Group By · Full step-by-step: Learner Guide, Lab 12


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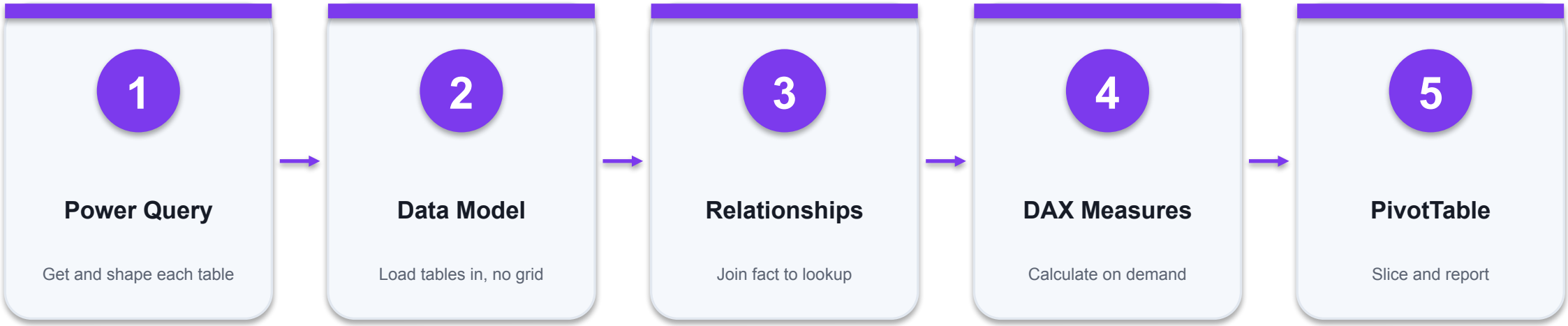
TOPIC 2 · POWER PIVOT

# Power Pivot is where the tables meet.

Power Query shapes each table; the Data Model relates them and DAX measures them.


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# Power Query and Power Pivot — the Division of Labour




**THE RULE**

Shape in Power Query. Relate and calculate in Power Pivot. Doing row-level clean-up in DAX, or trying to relate tables in M, fights the tools.


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# Star Schema — Fact and Lookup Tables

| Table | Kind | What it holds |
|---|---|---|
| **Sales** | Fact | One row per transaction — the numbers you measure |
| **Products** | Lookup / Dimension | One row per product — how you slice |
| **Calendar** | Lookup / Dimension | One row per date — time intelligence |

**WHEN IT MATTERS** Relationships run ONE-TO-MANY from the lookup table to the fact table. Lab 13 builds exactly this three-table model.


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# Implicit vs Explicit Measures

## ✓ Explicit measure (do this)

- Written once in the Calculation Area
- Total Sales := SUM(Sales[Amount])
- Reusable in every PivotTable and other measures
- Renaming it updates everywhere
- The professional practice

## ✗ Implicit measure (avoid)

- Created by dragging a field into Values
- Excel silently picks SUM for you
- Cannot be reused or referenced by name
- Disappears when the field is removed
- Fine for a quick look, wrong for a model

Always write explicit measures. Lab 13 creates four of them.


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# The Four DAX Measures You Build in Lab 13

Written in the Power Pivot Calculation Area, one measure per cell.

```dax
Total Sales := SUM(Sales[Amount])

Order Count := COUNTROWS(Sales)

Average Order :=
    DIVIDE([Total Sales], [Order Count])

Sales YTD :=
    TOTALYTD([Total Sales], Calendar[Date])
```

**SUM aggregates a column**

It respects whatever filters the PivotTable applies.

**COUNTROWS counts the table**

Not a column — so blank values still count as rows.

**DIVIDE, never /**

DIVIDE returns blank on a divide-by-zero instead of an error.

**TOTALYTD needs a date table**

Mark Calendar as a Date Table first, or time intelligence returns blank.


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TOPIC 2 · HANDS-ON LAB

LAB 13 · 30 MIN

# Lab 13: Build a Power Pivot Data Model with DAX Measures

The learner loads the fact and lookup tables to the Power Pivot Data Model, draws relationships in Diagram View, and writes explicit DAX measures.

**YOU'LL PRODUCE**

A Power Pivot data model with three related tables and four DAX measures.

**HOW YOU'LL KNOW IT WORKED**

A PivotTable with Products[Category] on rows and [Total Sales] in values returns per-category totals that sum to the grand total.

**MAPS TO** LO2 · A2 · K3, K4 — Load to the Data Model, create relationships and write DAX measures.

LAB 13

**lab-13-power-pivot-model.xlsx**

Excel, Power Query Editor, Power Pivot, DAX, Diagram View · Full step-by-step: Learner Guide, Lab 13


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# Topic 2 Recap

1. **M is case sensitive**
   if / then / else / and / or are always lowercase
2. **Every if needs an else**
   There is no optional third argument in M
3. **Date.AddDays**
   Derives deadlines that recalculate on refresh
4. **Unpivot Other Columns**
   The version that survives a new column
5. **Append = rows, Merge = columns**
   The most common mix-up
6. **Explicit DAX measures**
   Written once, reusable everywhere


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# Tea Break

15 minutes


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03

TOPIC 3

# Advanced Data Analysis using Power Query

The six joins · Merge · Approximate match · Fuzzy matching · Transformation tables


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TOPIC 3 · JOINS

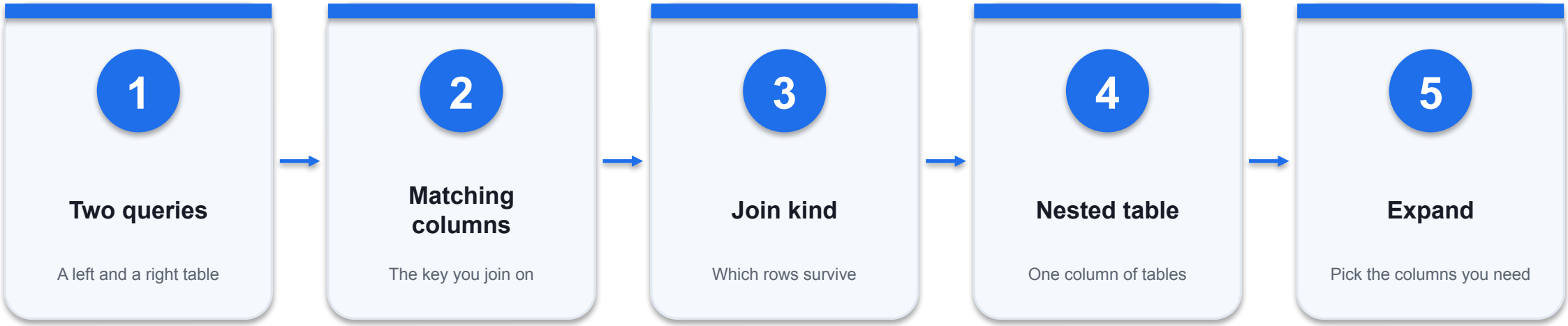
# Choosing the wrong join changes your row count silently.

No error, no warning — just a different answer. This topic makes the choice deliberate.


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TOPIC 3 · THE MERGE PIPELINE

# How a Merge Works




**THE STEP EVERYONE FORGETS**

A merge produces a COLUMN OF TABLES. Nothing useful appears until you click the expand arrow and choose which columns to bring through.


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TOPIC 3 · ALL SIX

# The Six Join Kinds

| Join kind | Which rows you keep | Typical use |
|---|---|---|
| **Left Outer** | ALL left rows + matches from the right | The safe default — an XLOOKUP replacement |
| **Right Outer** | ALL right rows + matches from the left | Rare — usually swap the tables instead |
| **Full Outer** | Every row from BOTH tables | Reconciling two systems |
| **Inner** | Only rows that match on BOTH sides | Confirmed, complete records only |
| **Left Anti** | Left rows with NO match on the right | Find orphans — orders with a bad product ID |
| **Right Anti** | Right rows with NO match on the left | Find unused lookups — products never sold |

**WHEN IT MATTERS** Lab 14 runs all six on the SAME pair of tables and records the row count each one returns — that comparison is your reference sheet for every merge.


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# Which Join Do I Need?




### Yes → Left Outer

Keeps all your records. Unmatched rows show null. The XLOOKUP equivalent.

### No → Inner or Anti

Inner keeps only matched rows. Left Anti keeps only the UNMATCHED ones.

A useful check: Inner rows + Left Anti rows = Left Outer rows. If your merge returns MORE rows than the left table, your key is not unique.


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TOPIC 3 · THE COMPARISON

# Merge vs XLOOKUP

## ✓ Merge (Power Query)

- Runs on Refresh, no dragging down
- Handles hundreds of thousands of rows
- Unmatched rows return null
- Can match on MULTIPLE columns
- Fuzzy matching available

## ✗ XLOOKUP (worksheet)

- Must be copied down as rows are added
- Slows the workbook as the range grows
- Unmatched rows return #N/A
- Needs a helper column for composite keys
- Exact or approximate only — no fuzzy

XLOOKUP is still the right tool for a one-off answer in a small sheet. Lab 15 replaces a column of XLOOKUPs with a single merge.


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TOPIC 3 · HANDS-ON LAB

LAB 14 · 30 MIN

# Lab 14: Walk Through All Six Join Kinds

Using one deliberately imperfect pair of tables — some orders reference a missing product, some products were never ordered — the learner runs all six join kinds in turn and records the row count each produces.

**YOU'LL PRODUCE**

Six merge queries and a completed row-count comparison table.

**HOW YOU'LL KNOW IT WORKED**

Left Outer returns exactly the Orders row count; Inner returns fewer; Left Anti plus Inner equals Left Outer.

**MAPS TO** LO3 · A3 · K2, K3 — Compare the six Power Query join kinds on the same pair of tables.

LAB 14

**lab-14-six-joins.xlsx**

Excel, Power Query Editor, Merge Queries · Full step-by-step: Learner Guide, Lab 14


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TOPIC 3 · HANDS-ON LAB

LAB 15 · 30 MIN

# Lab 15: Left Outer Join versus XLOOKUP

The learner replaces a worksheet full of XLOOKUP formulas with a single Left Outer merge, expands only the required column, and compares the two approaches on refresh behaviour, row-count safety and performance on a large table.

**YOU'LL PRODUCE**

A Sales query enriched with product details by merge instead of formulas.

**HOW YOU'LL KNOW IT WORKED**

The merged query has the same row count as Sales, and unmatched rows show null rather than #N/A.

**MAPS TO** LO3 · A3 · K2, K3 — Reproduce an XLOOKUP with a merge and expand only the needed column.

LAB 15

**lab-15-merge-vs-xlookup.xlsx**

Excel, Power Query Editor, Merge Queries, Expand · Full step-by-step: Learner Guide, Lab 15


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# Merging on More Than One Column

**When the key is a combination — product AND region — you must select both, in the same order, in both tables.**

### Ctrl-click to add

Click the first column, then Ctrl-click the second. The order is numbered.

### Order must match

Product-then-Region on the left must meet Product-then-Region on the right.

### Watch the row count

Matching on too few columns multiplies rows — Lab 16 shows a 5x inflation.


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LAB 16 · 30 MIN

# Lab 16: Merge on Multiple Fields

Prices vary by both product and region, so a single-column match returns the wrong price. The learner Ctrl-clicks two columns in each table to build a composite key, and learns that column selection order determines the pairing.

**YOU'LL PRODUCE**

A correctly priced sales query matched on Product plus Region.

**HOW YOU'LL KNOW IT WORKED**

Row count matches Sales exactly, and a North-region order carries the North price, not the South one.

**MAPS TO** LO3 · A3 · K2 — Match on a composite key using more than one column.

LAB 16

**lab-16-multi-field-merge.xlsx**

Excel, Power Query Editor, Merge Queries · Full step-by-step: Learner Guide, Lab 16


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TOPIC 3 · APPROXIMATE MATCH

# Power Query merges are EXACT match only.

VLOOKUP's approximate match must be rebuilt — with binning or a banding table.


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TOPIC 3 · BINNING

# Two Ways to Rebuild an Approximate Match

| Aspect | Conditional column (binning) | Banding table + merge |
|---|---|---|
| **How** | if/then/else on the thresholds | A table of Lower/Upper/Rate rows |
| **Bands live** | Inside the query | In a worksheet table anyone can edit |
| **To change a band** | Edit the query step | Edit a cell and Refresh |
| **Best for** | Two or three stable bands | Bands that change, or many of them |
| **Lab 17 builds** | Rate A — the quick version | The data-driven version |

**WHEN IT MATTERS** Both must agree at the boundaries. Lab 17 includes reps at exactly 10,000 and exactly 50,000 to prove it.


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LAB 17 · 30 MIN

# Lab 17: Approximate Match: Binning and Conditional Columns

Power Query merges are exact-match only, so VLOOKUP's TRUE behaviour must be rebuilt. The learner solves a commission-band lookup twice: once with a conditional column for fixed bands, and once with a banding table plus a merge so the bands stay data-driven.

**YOU'LL PRODUCE**

A commission-rate column produced by both the conditional-column and banding-table methods.

**HOW YOU'LL KNOW IT WORKED**

Both methods return identical rates, including at the exact band boundaries of 10000 and 50000.

**MAPS TO** LO3 · A3 · K2, K3 — Reproduce VLOOKUP's approximate match two ways.

LAB 17

**lab-17-approximate-match.xlsx**

Excel, Power Query Editor, Conditional Column, Merge Queries · Full step-by-step: Learner Guide, Lab 17


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# Cross Join and Drill Down

### 1 Cross join = every pair

Add a custom column whose formula is the other query, then expand. 5 regions x 13 products = 65 rows.

### 2 Why you want it

A report with no gaps — combinations with zero sales still appear, showing 0 instead of vanishing.

### 3 The danger

1,000 x 1,000 rows is a million-row table. Only cross join small lookup tables.

### 4 Drill Down

Right-click a single cell > Drill Down. The query now returns ONE value you can reuse as a parameter.

**LAB 18 COMBINES BOTH**

Build the complete region x product grid with a cross join, fill the gaps with zero, then drill down to a threshold value that drives a filter step.


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LAB 18 · 30 MIN

# Lab 18: Cross Join and Drill Down to a Variable

A sales matrix must show every region against every product, including combinations with no sales, so the report has no gaps. The learner builds that complete grid with a cross join, then uses Drill Down to turn a single cell into a reusable variable such as a target threshold.

**YOU'LL PRODUCE**

A complete region x product grid plus a drilled-down threshold variable.

**HOW YOU'LL KNOW IT WORKED**

The grid row count equals regions x products, and unsold combinations show 0 rather than being absent.

**MAPS TO** LO3 · A3 · K1, K2 — Build a complete grid with a cross join and extract a scalar by drill down.

LAB 18

**lab-18-cross-join-drilldown.xlsx**

Excel, Power Query Editor, Custom Column, Drill Down · Full step-by-step: Learner Guide, Lab 18


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TOPIC 3 · FUZZY MATCHING

# Real customer names never match exactly.

'Pte Ltd' vs 'Pte. Ltd.' vs a typo — fuzzy matching scores similarity instead of demanding equality.


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TOPIC 3 · THRESHOLD

# Tuning the Similarity Threshold

| Threshold | Behaviour | Risk |
|---|---|---|
| **1.00** | Exact match only — fuzzy is effectively off | Misses every variant |
| **0.90** | Very close spellings only | Still misses common abbreviations |
| **0.80** | The recommended starting point | Balanced — few false matches |
| **0.60** | Loosely similar text matches | False matches appear |
| **0.40** | Almost anything matches | Unusable — wrong data, silently |

**WHEN IT MATTERS** Lower the threshold and you catch more real matches but also more wrong ones. Start at 0.80, inspect the result, and adjust deliberately.


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# Fuzzy Matching Options

### 1 Similarity threshold

0 to 1. How alike two values must be before they pair. 0.80 is the sensible default.

### 2 Ignore case

Treats EVERGREEN FOODS and Evergreen Foods as the same.

### 3 Ignore space

Handles double spaces and stray padding.

### 4 Transformation table

A two-column From/To table teaching pairs no score can catch — 'ABC Group' means 'ABC Holdings'.

**WHY THE TRANSFORMATION TABLE IS ESSENTIAL**

No similarity score will ever link 'ABC Holdings' to 'ABC Group' — the words are genuinely different. Only an explicit mapping can. Lab 19 supplies one.


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LAB 19 · 30 MIN

# Lab 19: Fuzzy Matching and Transformation Tables

Customer names arrive spelled inconsistently across two systems — 'Pte Ltd' versus 'Pte. Ltd.', typos, and abbreviations.

**YOU'LL PRODUCE**

A reconciled customer query matched by fuzzy logic plus a transformation table.

**HOW YOU'LL KNOW IT WORKED**

All customers reconcile at threshold 0.8 with the transformation table applied, and no false match is introduced.

**MAPS TO** LO3 · A3 · K2, K3 — Match inconsistent text with fuzzy matching and a transformation table.

LAB 19

**lab-19-fuzzy-matching.xlsx**

Excel, Power Query Editor, Fuzzy Merge, Transformation Table · Full step-by-step: Learner Guide, Lab 19


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# Topic 3 Recap

1. **Six join kinds**
   Left/Right/Full Outer, Inner, Left/Right Anti

2. **Left Outer is the default**
   It keeps every row of your main table

3. **Anti joins find problems**
   Orphan records and unused lookups

4. **Composite keys**
   Ctrl-click both columns, same order, both tables

5. **Approximate match**
   Rebuild with binning or a banding table

6. **Fuzzy + transformation table**
   Score similarity, then teach the exceptions


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04

REVISION

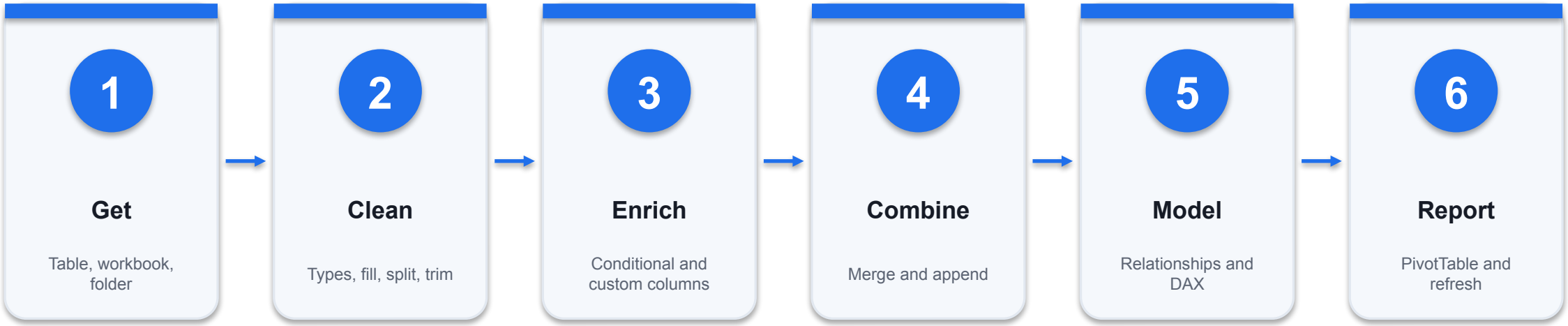
# Putting It All Together

The full pipeline, from raw file to answered business question


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REVISION · END TO END

# The Complete Workflow You Have Learned




**WHAT YOU CAN NOW DO**

Take a messy monthly extract and turn it into a refreshable, auditable report that rebuilds itself with one click next month.


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# The Pitfalls That Cost Most Marks

1. **Wrong data type**
   Set every type explicitly, first

2. **Capitalised If/Then**
   M keywords are always lowercase

3. **Missing else**
   Every if in M needs one

4. **Unpivot Only Selected**
   Use Unpivot Other Columns instead

5. **Merging on too few columns**
   Row count inflates silently

6. **Forgetting to expand**
   A merge yields a column of tables


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REVISION · RECOMMENDED COURSES

# Where to Go Next

1

## WSQ - Data Analytics with Excel

PivotTables, charts and dashboards on the clean data you now know how to build.

2

## WSQ - Statistical Data Analysis with Excel for Beginners

Add descriptive statistics and significance testing to your analysis.

3

## WSQ - Software Automation with Excel VBA Programming

Automate the steps Power Query cannot reach, with macros and VBA.

4

## WSQ - Create Interactive Dashboard Using Excel

Turn your Data Model into a slicer-driven interactive dashboard.


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ASSESSMENT

# Final Assessment

### 1 Written Assessment (WA)

Short-Answer Questions · 1 hour · open book.

### 2 Practical Performance (PP)

Hands-on Excel tasks · 1 hour · open book.

### 3 Open book means

Slides, Learner Guide and approved materials — no internet, no AI tools, no discussion.

### 4 Outcome

Competent / Not Yet Competent, with one re-assessment opportunity.

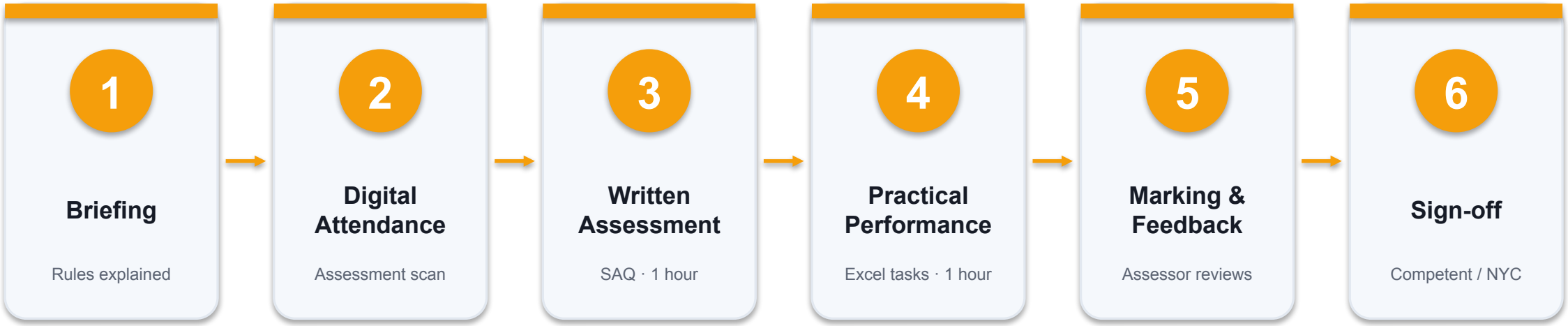
**BEFORE YOU START**

Clear your desk, silence your phone, and use a black or blue pen for the hard-copy script.


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ASSESSMENT

# Assessment Flow




**SIGN-OFF**

The assessor records the outcome, you sign the assessment record, and the result is released through the LMS/TMS portal.


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COURSE ADMIN

# Digital Attendance (Mandatory)

### 1 AM Attendance

Scan at the start of each training day.

### 2 PM Attendance

Scan after the lunch break.

### 3 Assessment Attendance

Scan before the final assessment begins.

### 4 75% Minimum

Attendance below 75% forfeits funding and certification.

The trainer will display the SSG digital attendance QR code — scan it with your phone camera and submit.


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COURSE CLOSING

# Certificate & TRAQOM Survey (Mandatory)

https://lms-tms.tertiaryinfotech.com

**Excel Power Query and Power Pivot**

- Trainer Slides (PPT)
- Learner Guide (PDF)
- Lesson Plan (PDF)
- Activities / Lab Files

**1 Log in to the portal**

Use your course account.

**2 Complete the TRAQOM survey**

Mandatory SSG course-quality feedback.

**3 Download your certificate**

Issued once assessed Competent.

**4 Keep the courseware**

Slides, guide and labs stay available to you.

The TRAQOM survey is mandatory for WSQ funded courses — please complete it before you leave.


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COURSE CLOSING

# Support

1

**Email**

enquiry@tertiaryinfotech.com

2

**Telephone**

+65 6100 0613

3

**Website**

www.tertiarycourses.com.sg

4

**LMS / TMS**

lms-tms.tertiaryinfotech.com


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# Thank You!

We hope Power Query and Power Pivot save you many hours a month.

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